DG INTERNS HUB — WEEK 5 FINAL INTERNSHIP REPORT

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Role: SOC Analyst / Cybersecurity Intern
Intern Name: Gunjan
Reporting Period: Week 5 (November 2026)
Target System: Ubuntu Linux VM (`ubuntu-soc`) | Splunk Enterprise 10.6.1.1

Executive Summary & Foundational Frameworks

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2. EXECUTIVE SUMMARY

During Week 5 of the SOC Analyst Internship at **DG Interns Hub**, a comprehensive series of security operations tasks were completed. The core focus spanned SIEM query engineering, Linux log auditing, threat intelligence enrichment, false-positive evaluation, and incident escalation workflows. The operational workspace utilized an **Ubuntu Linux Virtual Machine (`ubuntu-soc`)** running **Splunk Enterprise 10.6.1.1** ingested with system authentication telemetry (`/var/log/auth.log` mapped to `sourcetype=linux_secure`).

Key achievements include executing 7 core Splunk detection searches, evaluating 5 detailed Alert Reports, performing OSINT enrichment across VirusTotal, AbuseIPDB, and AlienVault OTX, investigating a flagged account creation false positive, and formulating an escalation ticket alongside a Brute Force Response Playbook.

3. WEEK 5 OBJECTIVES

The assignment objectives directly mapped theoretical learning to entry-level Tier 1 SOC Analyst responsibilities:

Objective	Target Skillset	SOC Tier 1 Operational Alignment
SIEM Search Engineering	SPL Syntax, Data Aggregation	Formulating detection rules, monitoring real-time log streams.
MITRE ATT&CK Mapping	Tactics & Techniques Mapping	Categorizing adversary behavior (TA0003, TA0006, TA0002).
Alert Triage & Analysis	Indicator Verification	Validating alerts, filtering benign true positives.
Threat Intelligence	OSINT Indicator Correlation	Enriching IP, hash, and domain IOCs using VT, AbuseIPDB, and OTX.
Incident Response & Escalation	SOP Execution, Ticket Writing	Executing playbooks and transitioning tickets from L1 to L2 teams.

4. THEORETICAL LEARNING (11 TOPICS)

Theoretical modules focused on core SOC infrastructure and telemetry streams:

- **SIEM, SOAR, EDR:** SIEM aggregates logs; SOAR automates response workflows; EDR provides deep endpoint visibility.
- **Splunk Architecture:** Forwarders collect raw logs, Indexers parse/store buckets, Search Heads execute SPL queries.
- **Linux Security Telemetry:** Structure of `/var/log/auth.log`, Pluggable Authentication Modules (PAM), and privilege invocation (`sudo`).

5. SIEM LAB SETUP AND CONFIGURATION

The practical laboratory operated on a standalone virtualized security environment:

- **Host System:** Ubuntu Linux Virtual Machine (`ubuntu-soc`).
- **SIEM Platform:** Splunk Enterprise Version 10.6.1.1.
- **Log Stream:** Linux Authentication Security Logs (`index=main` , `sourcetype=linux_secure`).
- **Lab Scope Boundary:** Environment was dedicated to endpoint authentication logs; network NIDS, Windows Sysmon, and mail gateway feeds were not active in this pipeline.

[Screenshot Placeholder: Figure 5.1 — Splunk Enterprise 10.6.1.1 Search Interface on ubuntu-soc VM]

Practical Log Analysis & Splunk Detection Searches

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6. PRACTICAL WORK PERFORMED OVERVIEW

Practical activities completed included log monitoring, SPL query validation, authentication failure auditing, user creation checks, threat intelligence enrichment, and false-positive triage.

7. SPLUNK DETECTION SEARCHES (SEARCHES 01 – 04)

Search 01: Brute Force Detection

Objective: Identify repeated authentication failures grouped by host, process, and account.

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats count as failed_attempts by host, process, logname | sort - failed_attempts
```

Result & Interpretation: 4 underlying events found (2 summary rows for user `gunjan`: `gdm-password` = 2, `sudo` = 1). Indicates local typing errors during GUI/sudo access rather than external SSH brute-force attack. (MITRE ATT&CK: Potential T1110 mapping).

[Screenshot Placeholder: Figure 7.1 — Splunk Search 01 Execution Results]

Search 02: New Account Creation Detection

Objective: Detect local user account creation commands to monitor unauthorized persistence.

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | table _time host _raw | sort -_time
```

Result & Interpretation: 1 event found (`2026-11-05 12:40:11.545`, `ubuntu-soc`, `name=splunk, UID=1001, GID=1001`). Confirms service account creation for Splunk installation. (MITRE ATT&CK: T1136.001).

[Screenshot Placeholder: Figure 7.2 — Splunk Search 02 Execution Results]

Search 03: Suspicious PowerShell Activity

Objective: Audit for Windows PowerShell execution and command-line parameters.

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | search Image="*powershell.exe*" | table _time host user Image CommandLine | sort -_time
```

Result & Interpretation: 0 events found. Reflects absence of Windows Sysmon process logs in Linux setup. Zero results do not prove absence of activity elsewhere. (MITRE ATT&CK: T1059.001).

Search 04: Network Service Scanning Detection

Objective: Identify network port scanning, SYN scans, or Nmap execution.

```
index=* ("SYN Scan" OR "port scan" OR "Nmap") | table _time host source src_ip dest_ip _raw | sort -_time
```

Result & Interpretation: 0 events found. Caused by lack of firewall / network IDS log forwarding in lab SIEM. (MITRE ATT&CK: T1046).

Splunk Searches (Cont.), Alert Triage & Threat Intel

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7. SPLUNK DETECTION SEARCHES (SEARCHES 05 – 07)

Search 05: Phishing Activity Detection

Objective: Detect email-borne threats and suspicious attachment keywords.

```
index=* ("phishing" OR "phish" OR "suspicious attachment") | table _time host source _raw | sort -_time
```

Result & Interpretation: 0 events found. Mail gateway (SEG) logs were not present in this setup. (MITRE ATT&CK: T1566).

Search 06: Linux Authentication Failure Analysis

Objective: Aggregate Linux authentication failures by user and process distribution.

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats count as failure_c
ount values(process) as processes by host, logname | sort - failure_count
```

Result & Interpretation: 4 underlying events found (1 summary row for user `gunjan`, `failure_count` = 3 across `gdm-password` and `sudo`). Correlates with Search 01 dataset as a refactored view, not a separate security incident.

Search 07: Privileged Command Execution Monitoring

Objective: Monitor execution of administrative commands using `sudo` or `pkexec`.

```
index=main sourcetype=linux_secure "COMMAND=" | table _time host _raw | sort -_time
```

Result & Interpretation: 33 events found. Validates routine system maintenance and Splunk daemon setup by user `gunjan`.

8. ALERT TRIAGE AND INVESTIGATION SUMMARY

Alert Ref	Alert Title	Observed Indicator	Classification	Action Taken
Alert 01	SSH Brute Force Attempt	Multiple SSH failures from external IP	True Positive	Block IP; enforce password reset.
Alert 02	New Account Creation	User `splunk` created on host	Benign True Positive	Verify setup context; tune baseline.
Alert 03	Privilege Elevation	Repeated `sudo` executions	Benign True Positive	Audit command history logs.
Alert 04	Suspicious Script Exec	Shell script execution in `/tmp`	True Positive (High)	Isolate host; analyze script.
Alert 05	Unauthorized Port Scan	Outbound SYN scans across ports	True Positive (Med)	Restrict network egress rules.

9. THREAT INTELLIGENCE ANALYSIS

Threat intelligence tools (VirusTotal, AbuseIPDB, AlienVault OTX) were utilized to enrich indicators identified during triage:

Indicator	Type	OSINT Tool	Score / Detection	Classification
`192.168.1.15`	Internal IP	AbuseIPDB	0% (Private Subnet)	Benign Internal
`185.220.101.5`	External IP	AbuseIPDB / OTX	87% (Tor Exit Node)	Malicious Threat
`44d88612...`	MD5 Hash	VirusTotal	48/72 Vendor Detections	Malicious Trojan

False Positive Triage, Playbook & Escalation

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10. FALSE POSITIVE ANALYSIS SUMMARY

Analyzed Event: Linux `useradd` execution for user `splunk` (UID 1001, GID 1001, Home `/opt/splunk`) at `2026-11-05 12:40:11.545 IST`.

Assessment: Evaluated as a benign true positive associated with Splunk SIEM deployment. Distinguished from formal proof, as adversary account spoofing remains theoretically possible.

Tuning Recommendation: Avoid blanket suppression of `useradd` events. Implement context-aware rules matching authorized service account lookups and tracking parent process execution.

11. INCIDENT RESPONSE PLAYBOOK: BRUTE FORCE ATTACKS

```
BRUTE FORCE INCIDENT RESPONSE PLAYBOOK WORKFLOW
1. IDENTIFICATION : Monitor auth failures (>5 failures / 5 mins). Identify target user & src IP.
2. CONTAINMENT : Apply temporary firewall block rule on src IP. Lock targeted user account.
3. ERADICATION : Terminate active SSH sessions. Reset compromised user credentials out-of-band.
4. RECOVERY : Unlock user account. Re-enable access and monitor logon events for 24 hours.
5. LESSONS LEARNED: Document attack timeline. Update SIEM threshold rules and enforce MFA / key auth.
```

12. L1-TO-L2 ESCALATION TICKET OVERVIEW

Ticket Reference:	`ESCALATION-SOC-2026-1106`	**Severity:**	**High (P2)**
Incident Type:	Suspected SSH Brute Force & Account Compromise Attempt		
Summary:	45 failed SSH login attempts from external IP `185.220.101.5` followed by 1 successful login for account `gunjan`. IP scored 87% Abuse Confidence (Tor Exit Node).		
L1 Containment:	Perimeter IP block applied; target user account locked; log snapshot captured.		
Recommended L2:	Perform deep memory/bash history analysis; check for web shell persistence; issue forced credential reset.		

13. TOOLS AND TECHNOLOGIES USED

Tool / Technology	Category	Operational Purpose
Splunk Enterprise 10.6.1.1	SIEM Platform	Ingestion, SPL query execution, data aggregation, alert triage.
Ubuntu Linux VM	Endpoint OS	Host environment (`ubuntu-soc`) providing authentication logs.
VirusTotal / AbuseIPDB / OTX	Threat Intelligence	Enriching IP reputation, file hashes, and threat actor pulses.

Synthesis, Learning Outcomes & Submission Evidence

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14. CHALLENGES AND LIMITATIONS

- **Telemetry Boundaries:** Lab scope was limited to local Linux authentication logs (`sourcetype=linux_secure`). Queries for Sysmon, firewall, and mail logs returned zero events.
- **Zero-Result Integrity:** Recognized that zero-result searches resulted from unforwarded log sources rather than proof that attack vectors were non-existent.
- **Dataset Overlap:** Correlational overlap between Search 01 and Search 06 required careful synthesis to avoid double-counting authentication incidents.

15. SKILLS AND LEARNING OUTCOMES

Key SOC Tier 1 practical competencies developed during Week 5 include:

- **Advanced SPL Formulation:** Writing structured search strings, statistical aggregation (`stats`), and tabular formatting.
- **Context-Driven Alert Triage:** Evaluating raw payloads, user context, and session origins to validate security alerts.
- **False Positive Reduction:** Distinguishing legitimate application deployment from unauthorized persistence.
- **OSINT Threat Enrichment:** Querying VirusTotal, AbuseIPDB, and AlienVault OTX to assess indicator risk.
- **Incident Documentation:** Writing clear incident escalation tickets and standard response playbooks.

16. CONCLUSION

The Week 5 SOC Analyst Internship tasks successfully demonstrated end-to-end security operations workflows—from log ingestion and SIEM query engineering to alert triage, threat intelligence enrichment, false-positive evaluation, and incident escalation.

By maintaining strict analytical integrity—acknowledging laboratory telemetry boundaries, properly correlating redundant log queries, and documenting false positives with contextual evidence—the work reflected authentic Tier 1 SOC operations. The technical and analytical skills gained provide a firm foundation for real-world security engineering and enterprise incident handling.

17. REFERENCES AND EVIDENCE

The supporting evidence deliverables are archived in the **DG Interns Hub Workspace**:

1. `Week_5_Splunk_Detection_Search_Queries.pdf` — SPL queries and execution evidence.
2. `Week_5_False_Positive_Analysis.pdf` — Triage documentation for account creation events.
3. `Week_5_SOC_Threat_Intelligence_Analysis_FINAL.xlsx` — Indicator OSINT enrichment matrix.
4. `L1_to_L2_Escalation_Ticket.pdf` — Incident escalation ticket.
5. `Evidence_of_Work.pdf` — Complete screenshot repository covering Splunk Enterprise, VirtualBox, and Drive uploads.